

# Digital Evidence Integrity Verification

1<sup>st</sup> Janani Rakshitha C.

CB.AI.U4AIM24009

*Artificial Intelligence(Medical Engineering)*

*Amrita Vishwa Vidyapeetham*

Coimbatore, India

cb.ai.u4aim24009@cb.students.amrita.edu

2<sup>nd</sup> Dasari Mohana Rakshiitha

CB.AI.U4AIM24010

*Artificial Intelligence(Medical Engineering)*

*Amrita Vishwa Vidyapeetham*

Coimbatore, India

cb.ai.u4aim24010@cb.students.amrita.edu

3<sup>rd</sup> Goudappolla Srikanth

CB.AI.U4AIM24011

*Artificial Intelligence(Medical Engineering)*

*Amrita Vishwa Vidyapeetham*

Coimbatore, India

cb.ai.u4aim24011@cb.students.amrita.edu

4<sup>th</sup> Hanumanthu Brahmini

CB.AI.U4AIM24053

*Artificial Intelligence(Medical Engineering)*

*Amrita Vishwa Vidyapeetham*

Coimbatore, India

cb.ai.u4aim24053@cb.students.amrita.edu

**Abstract**—The integrity of digital forensic evidence—such as DNA profiles, toxicology reports, and pathology findings—is critical in legal and medical domains. This project presents a blockchain-based system for verifying the authenticity of such digital files using SHA-256 hashing and Ethereum smart contracts. By storing hash values on a tamper-proof ledger, the system enables real-time validation of uploaded evidence without revealing its contents. A mismatch in hashes indicates potential tampering, thereby safeguarding evidence integrity and enhancing trust in digital forensic processes.

It eliminates the need for a central authority and provides a user-friendly interface for law enforcement, forensic experts, and medical professionals to verify the integrity of critical digital evidence in real-time.

**Index Terms**—Blockchain, digital evidence, integrity verification, SHA-256, Ethereum, Truffle, Ganache

## I. INTRODUCTION

In the modern era of digital forensics, electronic records such as DNA matching reports, toxicology analyses, and pathology results play a crucial role in legal investigations, criminal trials, and clinical diagnostics. These files often serve as critical evidence, and any compromise in their authenticity can lead to severe legal, ethical, or medical consequences. Traditional systems of storing and transferring such digital evidence typically rely on centralized databases or cloud services, which are susceptible to data tampering, unauthorized access, and breaches. Additionally, many existing systems do not offer robust mechanisms for verifying the originality of the evidence after it has been shared or archived.

The sensitivity of digital forensic evidence makes it imperative to implement solutions that not only store this data securely but also ensure its integrity over time. In conventional workflows, even minor alterations—intentional or accidental—may go undetected, potentially leading to wrongful convictions or compromised patient safety. Legal disputes surrounding manipulated or forged evidence further highlight the necessity for a trustless, transparent, and immutable veri-

fication mechanism. Without such a framework, stakeholders (including law enforcement agencies, legal professionals, and healthcare providers) face significant risk when relying on digital documents.

Many current solutions for digital evidence preservation and validation rely on centralized databases or third-party trust mechanisms. While systems like digital signature platforms and forensic audit trails provide a certain degree of authenticity validation, they do not entirely ensure the immutability and decentralization. Centralized authorities represent a single point of failure, meaning that these solutions can be compromised either from the inside or by an external cyberattack. Furthermore, many digital forensic tools are either proprietary or do not have interoperability, complicating sharing evidence across jurisdictions. In contrast, the proposed approach based on blockchain removes the necessity to trust intermediaries due to the Ethereum's decentralized nature. By storing only cryptographic hash of evidence, the system ensures the confidentiality of the data while providing tamper-proof publicly verifiable record of its integrity.

This project proposes a blockchain-based system to verify the integrity of digital forensic evidence using cryptographic hashing and Ethereum smart contracts. When a file is first uploaded, a SHA-256 hash of its contents is computed and stored immutably on the Ethereum blockchain via a smart contract. During future access or review, the system recomputes the hash and compares it against the blockchain-stored value. A match confirms the evidence is untampered, while any discrepancy indicates potential manipulation. The platform supports real-time verification without disclosing the actual contents of the file, thereby ensuring both confidentiality and trust. This solution significantly enhances the reliability of digital evidence management and provides a tamper-evident audit trail, fostering transparency and accountability in forensic and clinical domains.

## II. RELATED WORKS

### A. BMIAE (Blockchain-based Multi-instance Iris Authentication using ElGamal encryption)

Morampudi Mahesh Kumar, Munaga V. N. K. Prasad and U.S.N. Raju demonstrated how blockchain and smart contracts can preserve both integrity and privacy of biometric data. The authors highlight the use of homomorphic encryption (HE) to secure biometric templates. This encryption allows computations to be performed on encrypted data without needing to decrypt it first, thus maintaining confidentiality. The paper specifically utilizes ElGamal encryption for this purpose. Our work diverges from these by avoiding encryption, focusing only on hash-based verification of uploaded files to ensure integrity. BMIAE employs blockchain technology to enhance the integrity of the biometric templates and the results of the matching process. Smart contracts on the blockchain help ensure that the data remains tamper-proof and secure, addressing some of the limitations of traditional biometric systems.

To validate the effectiveness of BMIAE, the authors conducted experiments using several iris databases, including CASIA-V3-Interval, IITD, and SDUMLA-HMT. The results indicated that BMIAE not only improved accuracy but also eliminated the need to trust a centralized server, which is a significant advancement over existing methods.

### B. Privacy Preserving Fingerprint Authentication based on Homomorphic Encryption and Blockchain

Ali Zouaghi Yousra, Mahamdoua Meriama and Lahoulou Atidel presented a fingerprint authentication system that combines homomorphic encryption with Blockchain technology to ensure privacy and security. The proposed method allows for secure verification of fingerprints without exposing sensitive data. The authors discuss the advantages of using homomorphic encryption in biometric systems, particularly in terms of data confidentiality. The results show that the system is effective in preventing unauthorized access while maintaining high authentication accuracy. This work highlights the potential of advanced cryptographic techniques in enhancing biometric security.

### C. A Secure and Transparent Voting System using Blockchain Technology

Mrs. Anuradha Deokar et al. proposed a voting system that utilizes blockchain technology to establish a secure, transparent, and tamper-resistant platform for conducting elections. The primary goal is to mitigate the vulnerabilities inherent in traditional voting systems, such as manipulation, unauthorized access, and insufficient transparency. Additionally, the system enhances voter authentication by integrating Aadhar data for verification. The key components of the system include: blockchain integration, Aadhar based voter verification and smart contracts.

### D. Blockchain Technology in Supply Chain Management Enhancing Transparency and Efficiency

Sindhu Boianapalli et al. introduce effective solutions on how to integrate Blockchain with current systems through innovative hybrid systems, standardized user protocols, and new consensus protocols that optimize cost of implementation and environmental impact. The other aspect of the research undertakes to design clear regulatory frameworks and change management systems that can aid in the adoption of blockchain technology for traditional supply chain stakeholders. By utilizing real-world case studies and the simplification of smart contract deployment, this work illustrates the practical benefits blockchain can provide in enhancing supply chain operations. In conclusion, the goal of this study is to pioneer a sustainable, secure, and efficient blockchain ecosystem that promotes trust, transparency, and collaboration among supply chain partners, ensuring the future viability of the technology.

This research methodically proceeds the challenge of blockchain technology through a holistic approach in a multi-phase fashion to explore and improve the solution for global supply chains to remove these barriers with the transparency and efficiency and traceability. It aids in the identification of the shortcomings of the existing systems of using blockchain in supply networks and goes the extra mile to devise techniques that timely tackle those gaps and propel blockchain into the mainstream with the hottest supply chain systems. The goal of this study is to pioneer a sustainable, secure, and efficient blockchain ecosystem that promotes trust, transparency, and collaboration among supply chain partners, ensuring the future viability of the technology.

## III. METHODOLOGY

The project is a blockchain integrity verification system, which is constructed to ensure that any confidential digital forensic evidence, such as DNA match reports, toxicology analyses, and medical autopsy files, will not be tampered during communication and storage. The system is developed with the help of Ethereum smart contracts, SHA-256 hashing for file integrity, and a user-friendly frontend for prototype interaction. The implementation is carried out in several structured phases—smart contract development, blockchain setup and deployment, frontend integration, as well as file integrity verification.

### A. Smart Contract Design and Deployment

The application's main module resides in smart contract `EvidenceStore.sol`. It's written in Solidity language, and the code is compiled using Truffle. The contract offers two main functions: `uploadReport(bytes32 hash)` and `checkReport(bytes32 hash)`. The first one allows the user to record a hash of the file on Ethereum, and if the other report proves to be the same, the second function will return `TRUE`. In addition, the contract throws events after each successful upload for better traceability and transparency.

Truffle was chosen for the development framework due to its robustness in the compilation of Solidity contracts and

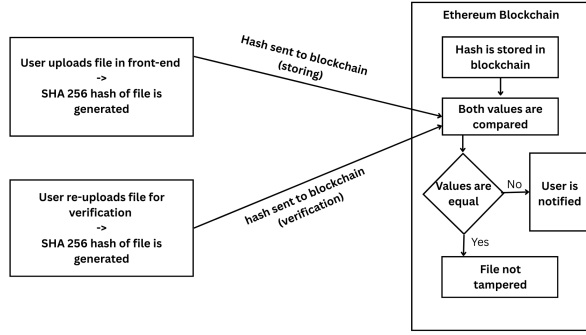


Fig. 1. Workflow

its effectiveness in deployment. On the other hand, Ganache was chosen for the local blockchain simulator, which made the development and testing process easy. After the contracts were compiled through Truffle, they were deployed to the Ganache blockchain and the corresponding contract address and ABI (Application Binary Interface) were obtained for interaction with the frontend.

#### B. Frontend Interface and Hashing Integration

- For developing the frontend of the project, we used pure HTML, CSS, and JavaScript in order not to use any other libraries or frameworks (like React) and keep the interface as simple and light as the task needed. For launching a local web server, we use the Live Server, which is an extension for Visual Studio Code. As for the frontend, it has only the following components: a file input, file hash, and blockchain interact buttons.
- A user selects a file, and JavaScript reads it and computes SHA-256 hash by using the Web API `crypto.subtle.digest`. The hash is the “digital fingerprint” of the file. The actual file contents are not uploaded to the blockchain or any cloud service; privacy is preserved, and no storage overhead is incurred. The hash generated is then sent to the blockchain by calling the `uploadReport()` function, or checked for its integrity by the `checkReport()` function.

#### C. Blockchain Communication and Metamask Integration

Web3.js was utilized to enable communication between the frontend and the Ethereum blockchain. The linking process was done through the ABI and contract address that was pulled from the compiled Contract. MetaMask, a browser-based wallet, was set up to connect to the local Ganache network with help of custom RPC URL and Chain ID (standard 1337). The Ganache Account keys were pulled into the MetaMask to replicate user transactions with a required test Ether of enough.

When MetaMask is connected, it can handle the signing of transactions and allocation of gas. When at the point of either uploading a document or verifying a user, the MetaMask pop-up will appear and require the user to approve the blockchain transactions. This way, we were able to ensure the

decentralized nature of the application, while making it even more user-friendly.

#### D. Integrity Verification Logic

The main aim of the project is to verify the integrity of digital forensic evidence through hash-based cryptography and decentralized storage. The core of the idea is that it is necessary to provide evidence that a digital file (for example, a DNA report or a toxicology report) has not been altered or otherwise modified since its initial upload. It is planned to use a two-stage mechanism to do this: SHA-256 hashing and querying of a smart contract.

File is selected by a user with the frontend interface. In the browser, FileReader API is used to read and convert this file into an ArrayBuffer. This buffer is an argument of the `crypto.subtle.digest()` method, which is available in the Web Crypto API. This file is then converted into a 256-bit-long fixed-length SHA-256 hash value using the SHA-256 hash. This process is entirely client-side and makes sure that the raw file is never leaked or sent over the network.

For further validation of the computed hash, it’s passed as a parameter to the `checkReport(bytes32 hash)` function of the deployed smart contract on the Ethereum blockchain. The smart contract holds a mapping of previously stored hash values, added using the `uploadReport()` function. Upon hash submission for validation, the function returns a boolean value: true if the hash exists (meaning the authenticity of the file and its originality are confirmed), or false if the hash doesn’t exist (meaning a potential mismatch or file tampering).

The approach guarantees that the tiniest modification in the file, like one character or byte, will lead to a hash that is entirely different, causing the integrity check to fail. Additionally, since the smart contract is run on a blockchain, the hash records are immutable and public, increasing the transparency and security of the integrity verification process.

## IV. RESULTS

TABLE I  
TRANSACTION COST FOR UPLOADREPORT() FUNCTION

| Files Uploaded | Gas Used | Cost (Gwei) | Cost (ETH) |
|----------------|----------|-------------|------------|
| 1              | 44438    | 133314      | 0.0001333  |
| 5              | 222190   | 666570      | 0.0006666  |
| 10             | 444380   | 1333140     | 0.0013331  |
| 15             | 666570   | 1999710     | 0.0019997  |

TABLE II  
EXECUTION COST FOR UPLOADREPORT() FUNCTION

| Files Uploaded | Gas Used | Cost (Gwei) | Cost (ETH) |
|----------------|----------|-------------|------------|
| 1              | 21962    | 65886       | 0.0000659  |
| 5              | 109810   | 329430      | 0.0003294  |
| 10             | 219620   | 658860      | 0.0006589  |
| 15             | 329430   | 988290      | 0.0009883  |

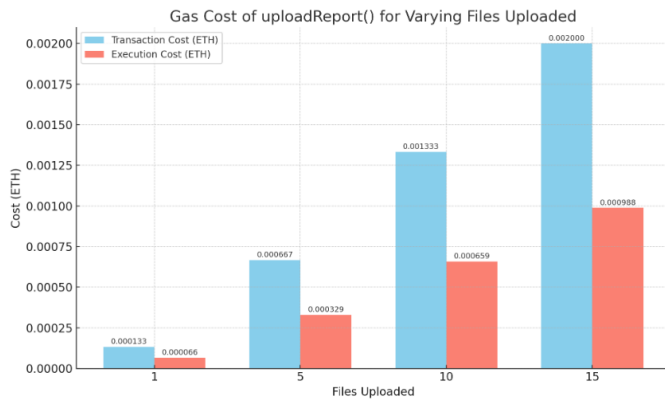


Fig. 2. Transaction gas cost and Execution gas cost comparison for uploaded files

## V. CONCLUSION

In this project, a decentralized method for integrity control of digital forensics evidence has been demonstrated. The system is based on Ethereum smart contracts and cryptographic hashing. The SHA-256 algorithm and the blockchain storage make it possible to detect document manipulation with a high probability. This is useful for identifying manipulation in such documents as DNA matching reports, toxicology results, and other forensic documents. The solution allows the user to upload files and check them with no transmission of the actual document. The system interface allows the user to upload and check documents without ever transmitting the actual files.

This method tackles the main problems of the old-fashioned digital evidence management systems like vulnerabilities in the center, unauthorized entrance, and the risk of unnoticed modifications. Using blockchains transparency and tamper resistance, it sets the base for the next secure digital asset recording in the law enforcement, healthcare, and legal sectors.

The prototype constructed affirms that the idea of integrity verification through blockchain is indeed possible. It also lays a foundation for applying it in different real-life applications where digital trust and traceability are required.

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